AI-Powered E-Grievance Hub For Smart Complaint Management

A

Seminar

Submitted in partial fulfilment for the award of the degree of

BACHELORS OF TECHNOLOGY

IN

Computer Science Engineering

by

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Certificate from Company

CERTIFICATE

This is to certify that the work titled "AI-Powered E-Grievance Hub For Smart Complaint Management" submitted by "V. Anil Reddy (BETN1CS173), S. Sai Pragathi (BETN1CS22101)" in partial fulfilment for the award of the degree of B.Tech. (CSE), ITM University, Gwalior has been carried out under my/our supervision.
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Designation

Date

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(Signature of the candidates)

V. Anil Reddy
S. Sai Pragathi

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Acknowledgement

Firstly, We thank Lord Almighty for making it possible for us to complete this work. The success and outcome of this project required a lot of guidance and assistance from many people, and we are incredibly privileged to have got it all along with the completion of my project. All that we have done is only due to such supervision and assistance and we would not forget to thank them.

I respect and thank Ms. Gauri Sharma for providing us an opportunity to do the project work and giving us all support and guidance which made us complete the project duly. She took keen interest on our project work and guided us all along and provided all the necessary information for developing a good system.

I would not forget to remember HOD, Mentors, DEAN etc....for their encouragement and more over for his timely support and guidance till the com- pletion of our project work. I heartily thank our Ms. Gauri Sharma for their guidance and suggestions during this project work.

At last but not the least we would like to thank our parents who trusted us and helped us through the errand.

Abstract

Grievance handling is an important part of any educational institution, yet many students still struggle with slow responses, unclear procedures, and the absence of proper tracking once a complaint is submitted. Most of the existing systems rely on manual methods, which often lead to delays and miscommunication. To address these issues, this project introduces E- Grievance Hub, a web-based platform that helps students raise their concerns easily and provides a clear structure for administrators to respond. The system is built using the MERN stack and allows students to submit grievances, attach documents, and follow the progress of their complaint through a transparent timeline. Administrators can review cases, update the status, and take necessary actions within the same platform. The use of role-based access ensures that each user interacts only with the features related to their responsibilities. An optional AI component is also included to help categorize complaints and direct them to the right department, reducing unnecessary delays. Overall, this project aims to create a simple, accessible, and accountable grievance management system that improves communication between students and the administration. By reducing manual work and bringing all interactions into a single digital space, the platform helps institutions handle grievances more efficiently and with greater transparency.

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List of Abbreviations

CSE – Computer Science Engineering HOD – Head of department

DFAT- Digital Forensica Automate Tool

dd- Disk Dump

JPEG- Joint Photographic Experts Group

HTML- Hyper Text Markup Language PNG- Portable Network Graphics

PDF- Portable Document Format

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Chapter 1

Introduction

Every educational institution deals with a wide range of student concerns, whether they are related to academics, facilities, administration, or day-to-day campus experiences. While these issues are common, the way they are handled often becomes slow and confusing for students. Many colleges still depend on traditional paper- based or in-person complaint procedures, which makes it difficult for students to raise concerns and even harder for them to keep track of what happens afterward. As a result, many grievances remain unresolved or are addressed much later than expected.

With increasing digitization in all sectors, there is a growing need for a simple and reliable platform that allows students and administrators to communicate more effectively. This project, E-Grievance Hub, is designed to fill that gap by offering a structured and user-friendly system for grievance submission and management. Instead of waiting in queues or relying on manual follow-ups, students can quickly file a grievance online, attach any supporting documents, and monitor its progress in real time.

The system also supports administrators by giving them a clear workflow for reviewing, updating, and resolving grievances. Everything is neatly documented, which helps maintain transparency and reduces misunderstandings. By bringing the entire process onto a single digital platform, the project aims to create smoother communication, faster resolutions, and a more accountable grievance redressal environment within the institution.

Objectives:

The main aim of this project is to create a smooth and transparent system that helps students submit their grievances easily and allows administrators to handle them in an organized way. Instead of depending on manual processes, the goal is to build a digital platform that makes communication faster and more reliable. The following objectives guide the development of the E-Grievance Hub:

1. To provide a simple platform for students to submit grievances Many students struggle with complicated procedures, so the first objective is to offer a clean and user-friendly interface where they can file complaints without confusion.

2. To allow students to track their complaint status A major goal is to make the grievance process transparent by letting students see updates, actions taken, and the current status of their complaint at any time.

3. To help administrators manage grievances efficiently The system aims to reduce manual workload by giving admins a proper dashboard where they can view, update, and respond to grievances based on priority and department.

4. To ensure secure and role-based access Another key objective is to make sure that only authorized users (Student, Admin, SuperAdmin) can access the features meant for them, improving both security and accountability.

5. To support document uploads and proper evidence handling Students should be able to attach screenshots, PDFs, or images so that admins can understand the issue clearly and respond more effectively.

6. To provide clear communication between students and administrators The objective here is to avoid miscommunication by giving both sides a structured way to exchange updates, comments, and official responses.

7. To maintain a complete digital record of all grievances The system should store every complaint, status change, and action taken, enabling institutions to view past cases and generate reports when needed.

8. To make grievance handling faster, organized, and more transparent Overall, the main objective is to build a platform that eliminates delays, reduces confusion, and makes the entire grievance process more trustworthy and easier for everyone involved.

Problem Statement:

In most colleges, the process of submitting and resolving student grievances is still slow, confusing, and mostly dependent on manual work. Students often have to write applications, visit offices, or follow multiple people just to know what happened to their complaint. Because there is no proper tracking system, many grievances end up being delayed, misplaced, or without any clear communication. Students also have no way to see who is handling their issue or how far it has progressed. On the other side, administrators also face difficulties because they have to manage complaints manually, maintain records, and respond to students without a centralized system.

This leads to miscommunication, extra workload, and a lack of accountability. Different departments handle complaints differently, which creates inconsistency and further delays. Due to all these issues, there is a clear need for a simple, transparent, and organized grievance management platform that allows students to raise concerns easily and helps administrators track and resolve them efficiently.

A system that provides proper status updates, a clear workflow, document attachments, and department-wise routing can make the entire process smoother and more reliable. The E-Grievance Hub aims to solve these problems by bringing everything into one digital platform where both students and administrators can communicate more effectively and resolve issues without confusion or delay.

Chapter 2

Literature Review :

S.No	Title	Journal	Author	Limitations
1	Effectiveness of Online Grievance Redressal and Management System (iGRAM)	Indian Journal of Educational Technology	Upendra Nabh Tripathi, Amit K. Srivastava, Bhanu Pratap Singh NCERT Journals	Limited to IGNOU distance learners; no technical architecture details; suggestions limited (mobile app, popularization)
2	Effectiveness of Grievance Redressal Mechanism: Students' Perceptions	International Journal of Social Sciences Bulletin	Alveena Javed, Jamil Chitrali ijssbulletin.com	Focuses on perception rather than system design; lacks implementation detail; purely survey-based ijssbulletin.com
3	State-Level Students Grievance Support System	International Journal of Scientific Research in Computer Science, Engineering and IT	Satheeswaran Venkatesan, Arjun R, Nidhin A, Pranav C.N2 ijsrcseit.com	Paper describes prototype; doesn't include role-based access control, timeline tracking or AI; limited evaluation.
4	Addressing Student Grievances: A Comprehensive Analysis of Grievance Management Systems	International Journal of Science, Engineering and Technology	Harshil Sheth, Siddhraj Solanki, Aayush Parmar, Jay Dave, Hiren Raithatha IJSET	Conceptual / survey-based; lacks deep implementation or evaluation; no AI module or detailed architecture.

5	An Online Grievance Redressal System	International Research Journal of Engineering and Technology (IRJET)	Mukesh Buldak, Shrikant Pandhekar, Afzal Gigani, Amreshsinh Kachwah, Kundan Patil, Poonam Polshetwar, Pradeep Jadhav studylib.net	Very basic CRUD system; no analytics, no escalation, no timeline; not very scalable.
6	Very basic CRUD system; no analytics, no escalation, no timeline; not very scalable.	International Journal of Scientific Research in Engineering and Management	Chandresh Bind, Chirag Goel, Ayush Rai ResearchGate	Strategic / theoretical analysis; doesn't provide technical implementation; limited empirical data
7	Student Grievance Redressal Cell	International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)	Nitin Janwe, Srushti Awari, Gaytri Burabde, Pratikasha Kande IJARSCT	Simple web app; basic role flow (student and clerk); lacks analytics, AI categorization, and detailed workflow

Reviews :

1. When I read the paper on the IGNOU iGRAM grievance system, it helped me understand how large-scale educational institutions manage complaints in a fully digital way. What stood out to me was how they focused more on reducing delays and improving communication between students and the authorities, which is something I strongly relate to because our institution faces similar issues. I liked how the system tries to make students feel heard and ensures that no complaint is lost or ignored. But while reading it, I also felt that the paper mainly talks about the outcomes and overall effectiveness rather than the technical details. It does not explain how the backend works. This made me realize that even though real-world examples are very helpful to understand the problem space, I still need to design the core architecture on my own. The paper motivated me to build a system that is transparent and

user-friendly like iGRAM, but with better technical depth, proper timelines, and more structured admin controls.

2. This paper really opened my eyes to the softer side of grievance systems — how students feel, how they react, and what stops them from raising complaints. Many times, we think of grievances as just "tickets" or "requests," but the study made me realize that they carry emotions too, like worry, fear, stress, or frustration. The authors explained that some students hesitate to submit complaints simply because they do not trust the system or they don't understand the process well. This influenced my thinking a lot. I realized that a system is not helpful unless students are actually comfortable using it, so I focused on making the UI simple and clear with proper guidance on every step. Although the paper doesn't include technical components or architecture, its insight into student psychology helped me design my platform with transparency and friendliness in mind. This made the student portal of my project feel more welcoming and easy to use rather than complicated or intimidating.

3. The prototype presented in this paper gave me a good reference for how a digital grievance platform could work at a state level. Their system follows a simple structure where students can register complaints, add documents, and receive updates. It was useful for me because it gave a clean idea of how the basic cycle of a grievance system works. But as I studied the paper more deeply, I realized it lacked many features that a modern system should have. There was no support for escalation, no proper admin hierarchy, no timeline feature, and no automated classification or routing. It became clear that the system might work for a small environment, but it would struggle to scale or handle real institutional complexities. This helped me think critically about my own design. I realized I needed to add multi-role access (Student, Admin, SuperAdmin), detailed timeline history, proper notification system, and more structured workflows. So even though the paper was helpful as a starting point, I ended up building a much more advanced and polished system.

4. This paper helped me understand the typical problems that occur in college grievance systems and the wide variety of complaints students bring up. Instead of focusing on technical development, the authors analyzed the entire grievance environment, which gave me a new perspective on what issues really matter to students. They mentioned delays, unclear procedures, and poor communication — all of which are common issues in many institutions. Reading this made me think deeply about what features my project should include, not from a technical lens but from a management point of view. The idea of categorizing grievances properly and providing consistent updates came directly from this paper. Since it didn't go into coding or system architecture, I had to figure out how to

convert these management concepts into working features. This is why I added modules like category selection, timeline updates, proper document uploads, and department-wise routing. The paper helped me understand the real operational needs behind a grievance system, and I built my features around that insight.

5. The IRJET paper was one of the simplest systems I reviewed, but it gave me a clear idea of how the basic workflow is structured in a digital grievance system. Their model shows how students register, login, submit grievances, and check the status. This helped me visualize the initial structure of my system. However, as I continued reading, I realized that their solution is very limited. There was no support for multiple departments, no admin roles beyond a basic reviewer, no escalation system, and no detailed complaint history. It also didn't include modern concepts like AI categorization or automated routing. Because of these limitations, I treated this paper more like a beginner-level reference. It showed me what the minimal requirements are, but it also pushed me to build a system that goes far beyond that. I made sure to include multi-level roles, secure authentication, proper admin dashboards, and real-time notifications so that my system addresses all the shortcomings that I noticed in their design.

6. This paper helped me understand the overall structure and philosophy of how grievance systems should function within institutions. It goes into detail about transparency, fairness, accountability, and proper communication between the administration and the students. Even though it does not provide technical details or any coding structure, it helped me understand the policy-level foundation behind a grievance system. I realized that building the software is only half the job; the system also needs to follow proper institutional norms. Because of this, I designed features like escalation paths, status justification, and proper documentation for every action taken by the admin. These features make my system not just a technical project but also an institution-ready model. The paper also made me think about what happens behind the scenes after a grievance is submitted, such as committee review and follow-up, which inspired me to structure my admin modules more carefully.

7. This paper focuses more on improving the student experience and learning environment by solving grievances in a structured way. I found it helpful because it explained real challenges like miscommunication, delays, and lack of documentation. These issues are very common in many colleges and something I have personally observed too. Although the paper is not very technical, it does highlight the need for a proper system that is easy to use and ensures grievances do not get lost. That made me think about simplicity and reliability when designing my own platform.

Chapter 3

Project details

Methodology :

The development of the E-Grievance Hub was carried out in a structured way so that each part of the system was built clearly and efficiently. The methodology mainly focuses on three core areas: the frontend, the backend, and the algorithms/technologies used to implement different features. All these parts were developed step-by-step to make the platform simple, secure, and reliable for both students and administrators.

1. Frontend Methodology The frontend of the system was designed to be as simple and user-friendly as possible because the main users are students and staff who should not have difficulty navigating the platform. I used React.js because it allows building flexible and reusable components, which helped keep the UI clean and organized.

1.1 Technologies Used

• React.js – for building UI components • React Router – for page navigation • Context API – for managing user login state • Axios – for sending API requests • CSS – for styling and maintaining clean layouts • FormData – to handle file uploads like PDFs or images

1.2 Frontend Workflow

1. User logs in or signs up 2. Based on the role (Student/Admin/SuperAdmin), the system redirects to the correct dashboard 3. All API requests are sent through a single axios instance 4. Pages display real-time status, grievance details, and notifications 5. Students can upload files using FormData 6. Admin pages show advanced options like status updates, assigning grievances, and viewing timelines

1.3 Main Frontend Components Developed

• Signup & Login pages • OTP verification • Student dashboard • Grievance submission form • View grievances • Timeline tracking page • Admin dashboard • Admin grievance details page • SuperAdmin dashboard The frontend was developed to ensure clarity, easy navigation, and minimal steps for filing or managing grievances.

2. Backend Methodology The backend handles all business logic, data storage, authentication, and communication with the frontend. I used Node.js and Express.js to build a modular backend with separate controllers and routes for each feature.

2.1 Technologies Used • Node.js – server-side runtime • Express.js – to build API routes

• MongoDB – database to store users, grievances, departments, etc. • Mongoose – object modeling for MongoDB • JWT (JSON Web Tokens) – secure authentication • Multer – for uploading files (images, screenshots, PDFs) • BCrypt – for password encryption • Nodemailer – for sending email notifications

2.2 Backend Workflow 1. User registers → OTP verification (optional) 2. Login generates a JWT token 3. Student submits grievance → stored in MongoDB 4. Admin views assigned grievances from the dashboard 5. Admin updates status → triggers notifications 6. Timeline gets updated automatically 7. SuperAdmin approves or rejects admins 8. Database stores every action for transparency and records

2.3 Backend Modules Implemented • Auth Module – registration, login, password reset • User Module – student profile, OTP verification • Grievance Module – submission, status update, escalation, timeline • Complaint Type Module – create, update, and delete types • Notification Module – alerts for updates and actions • SuperAdmin Module – admin approval, department management • Upload Module – handling documents and proof files The backend was made scalable so new departments, admins, or features can be added easily in the future.

3. Algorithms / Technologies Used To ensure security, fast communication, and proper workflow, several algorithms and technical methods were used:

3.1 JWT Authentication Algorithm Used to securely authenticate students, admins, and super admins. Purpose: • Protects routes from unauthorized users • Allows role-based access • Keeps login sessions safe

3.2 BCrypt Hashing Algorithm Every password is encrypted before storing so that even if the database is exposed, passwords cannot be read. Purpose: • Improves security • Prevents plain-text passwords • Protects user identity

3.3 Multer File Upload Algorithm Used to handle multipart/form-data for grievance attachments. Purpose: • Supports uploading images, PDFs, screenshots • Validates file size and format • Saves files securely

3.4 Role-Based Access Control (RBAC) Controls what each type of user can do: • Student: file grievances, view timeline • Admin: update status, assign, comment, resolve • SuperAdmin: approve admins, manage departments Purpose: • Ensures workflow discipline • Prevents misuse of features • Keeps data access secure

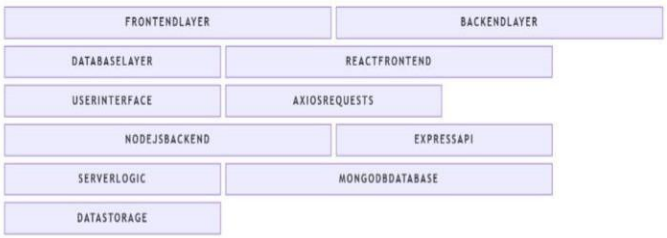
3.5 Optional AI Classification (If implemented)

Basic ML algorithms like: • Naïve Bayes • SVM • Logistic Regression can automatically classify grievances into categories. Purpose: • Helps assign complaints faster • Reduces admin workload Implementation : → System Architecture: The system architecture of the E-Grievance Hub follows a three-layer structure: the frontend, the backend, and the database. All three layers communicate with each other to provide a smooth experience for students, admins, and the super admin.

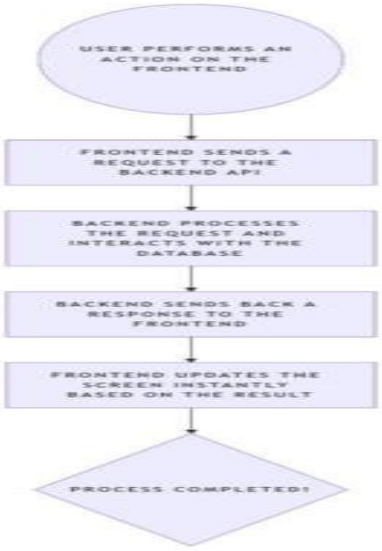
1.1 Frontend Layer (Client Side) This is the part of the system that users interact with. It includes pages like: • Login / Signup • Student Dashboard • File Grievance Page • Admin Dashboard • SuperAdmin Dashboard The frontend is built using React.js, which helps create reusable components and a clean user interface. It sends requests to the backend using Axios and receives responses in real time.

1.2 Backend Layer (Server Side) The backend is responsible for all logic and operations. It handles: • User authentication • Grievance submission • Status updates • Notifications • File uploads • Admin approvals The backend is built using Node.js and Express.js, and it exposes APIs that the frontend can call. Every action taken on the frontend (like submitting a grievance) goes through the backend.

1.3 Database Layer The database stores all the information permanently, such as: • Student details • Admin information • Grievances • Departments • Complaint types • Notifications MongoDB is used because it stores data in a flexible JSON-like format, making it easier to handle complex objects like images, grievance history, and attachments.



Communication Flow :
Here is how everything communicates:



Module Explanation :
The system is divided into several modules so that each function can be developed, tested, and maintained separately. Below is the explanation of each module in very simple terms.

2.1 Authentication Module This module handles all login and signup-related tasks. What it does: • Registers students and admins • Verifies OTP (if required) • Handles login with JWT authentication • Protects routes using tokens • Allows password reset This ensures that only valid and verified users can enter the system.

2.2 User Module (Students) This module contains everything related to students. What it does: • Lets students create grievances • Upload documents or screenshots • View their grievance history • Track the timeline of each complaint • Receive notifications It ensures that students have a simple and clear experience.

2.3 Admin Module The admin module is used by college staff who handle the grievances. What it does: • Shows all assigned grievances • Allows updating the status (Pending → In Progress → Resolved) • Add comments and notes • Upload response documents • Forward or escalate grievances • View student details Admins handle the actual resolution process.

2.4 SuperAdmin Module The SuperAdmin has the highest level of control in the system. What it does: • Approves or rejects admins • Manages all departments • Views analytics and overall reports • Monitors system activity This module ensures proper administrative control and security.

2.5 Grievance Module This is the core module of the entire system. What it does: • Stores student grievances • Assigns them to departments • Tracks every update •

Maintains timeline history • Supports attachments • Escalates grievances if needed It manages the complete life cycle of a grievance.

2.6 Notification Module This module increases communication between students and admins. What it does: • Sends email alerts • Generates in-app notifications • Marks notifications as read/unread • Alerts students when status changes This helps users stay updated without checking repeatedly.

2.7 Complaint Type & Department Module This module lets admins and SuperAdmins organize grievances properly. What it does: • Add and remove complaint types • Manage department list • Assign grievances to the correct category It improves clarity and routing accuracy.

2.8 File Upload Module This module handles all documents and screenshots. What it does: • Accepts PDF, JPG, PNG files • Validates file size • Stores files securely • Connects uploaded files with grievances This is used when students submit proof or admins upload responses.

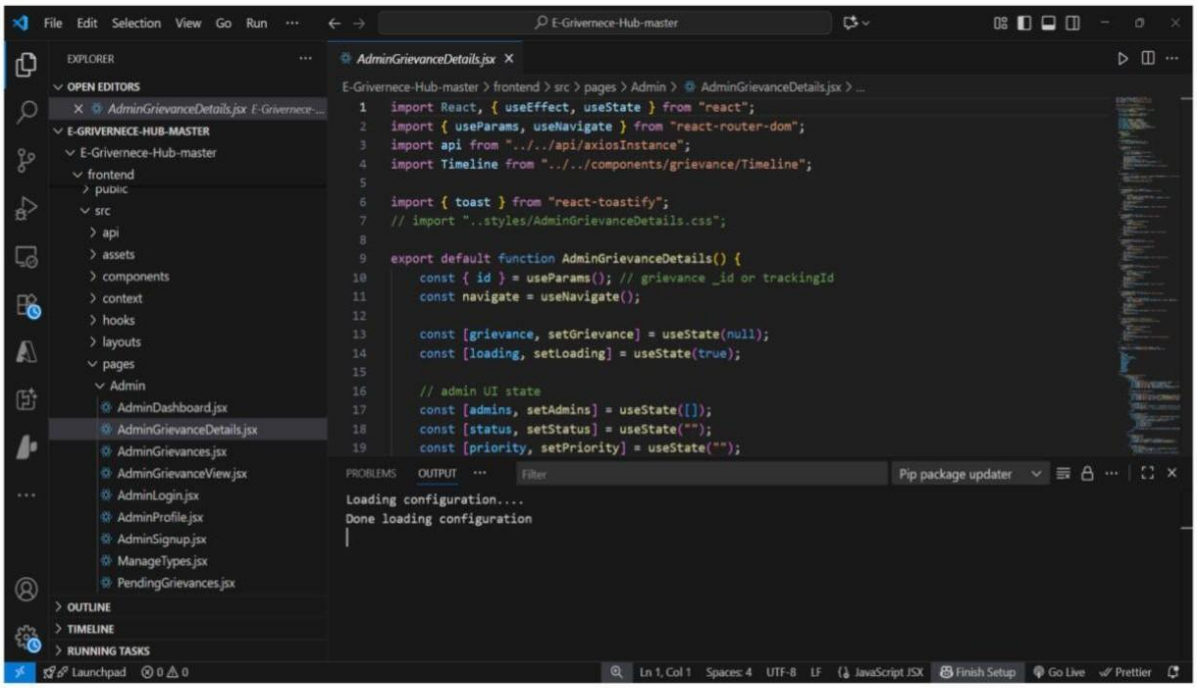
Screenshots/Diagrams :

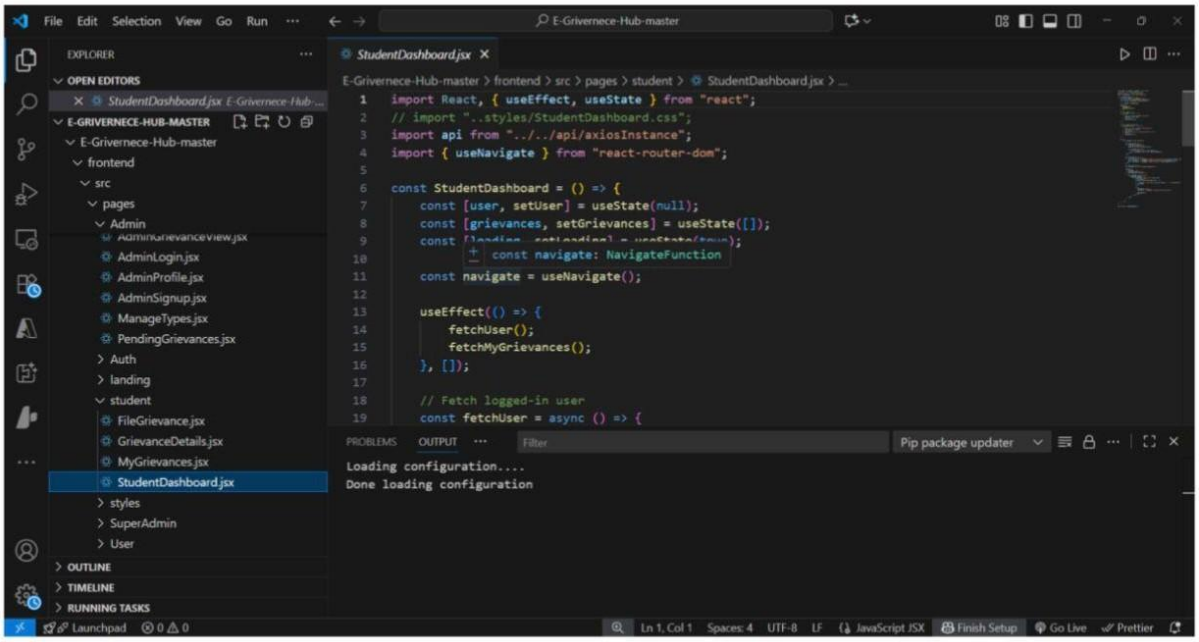
```
# *FRONTEND STRUCTURE (FINAL)*

src/
│
├── api/
│   └── axiosInstance.js
│
├── assets/
│   ├── images/
│   └── icons/
│
├── components/
│   ├── common/
│   │   ├── Navbar.jsx
│   │   ├── Footer.jsx
│   │   ├── Loader.jsx
│   │   └── NotificationDropdown.jsx
│   ├── grievance/
│   │   ├── ImageSlider.jsx
│   │   └── Timeline.jsx
│   └── protected/
│       └── ProtectedRoute.jsx
│
├── context/
│   ├── AuthContext.jsx
│   └── UserContext.jsx
│
├── hooks/
│   └── useAuth.js
│
├── layouts/
│   ├── UserLayout.jsx
│   ├── AdminLayout.jsx
│   └── SuperAdminLayout.jsx
│
├── pages/
│   ├── Landing/
│   │   └── LandingPage.jsx
│
│   ├── Auth/
│   │   ├── Signup.jsx
│   │   ├── VerifyOtp.jsx
│   │   └── Login.jsx
│
│   ├── User/
│   │   ├── Dashboard.jsx
│   │   ├── CreateGrievance.jsx
│   │   ├── MyGrievances.jsx
│   │   └── TrackGrievance.jsx
│
│   ├── Admin/
│   │   ├── AdminDashboard.jsx
│   │   ├── AdminLogin.jsx
│   │   ├── AdminSignup.jsx
│   │   ├── AdminProfile.jsx
│   │   ├── AdminGrievances.jsx
│   │   ├── AdminGrievanceView.jsx
│   │   ├── AdminGrievanceDetails.jsx
│   │   ├── ManageTypes.jsx
│   │   └── PendingGrievances.jsx
│
```

BACKEND STRUCTURE (FINAL)

```
backend/
│
├── server.js
├── .env
│
├── Database/
│   └── ConnectDB.js
│
├── controllers/
│   ├── adminController.js
│   ├── authController.js
│   ├── complaintType/
│   │   ├── createType.js
│   │   ├── deleteType.js
│   │   └── getTypes.js
│   ├── grievanceController.js
│   ├── notificationController.js
│   └── userController.js
│
├── middleware/
│   ├── authMiddleware.js
│   ├── errorHandler.js
│   └── upload.js
│
├── models/
│   ├── Admin.js
│   ├── ComplaintType.js
│   ├── Department.js
│   ├── Grievance.js
│   ├── Notification.js
│   └── User.js
│
├── routes/
│   ├── adminRoutes.js
│   ├── authRoutes.js
│   ├── complaintTypeRoutes.js
│   ├── grievanceRoutes.js
│   ├── notificationRoutes.js
│   ├── superAdminRoutes.js
│   └── userRoutes.js
│
└── utils/
    ├── generateToken.js
    └── sendEmail.js
```


File Edit Selection View Go Run
E-Grivernece-Hub-master
EXPLORER
OPEN EDITORS
AdminGrievanceDetails.jsx E-Grivernece-...
E-GRIVERNECE-HUB-MASTER
E-Grivernece-Hub-master
frontend
public
src
api
assets
components
context
hooks
layouts
pages
Admin
AdminDashboard.jsx
AdminGrievanceDetails.jsx
AdminGrievances.jsx
AdminGrievanceView.jsx
AdminLogin.jsx
AdminProfile.jsx
AdminSignup.jsx
ManageTypes.jsx
PendingGrievances.jsx
OUTLINE
TIMELINE
RUNNING TASKS
AdminGrievanceDetails.jsx
E-Grivernece-Hub-master > frontend > src > pages > Admin > AdminGrievanceDetails.jsx > ...
import React, { useEffect, useState } from "react";
import { useParams, useNavigate } from "react-router-dom";
import api from "../../api/axiosInstance";
import Timeline from "../../components/grievance/Timeline";
import { toast } from "react-toastify";
// import "..styles/AdminGrievanceDetails.css";
export default function AdminGrievanceDetails() {
const { id } = useParams(); // grievance _id or trackingId
const navigate = useNavigate();
const [grievance, setGrievance] = useState(null);
const [loading, setLoading] = useState(true);
// admin UI state
const [admins, setAdmins] = useState([]);
const [status, setStatus] = useState("");
const [priority, setPriority] = useState("");
PROBLEMS OUTPUT
Filter
Pip package updater
Loading configuration....
Done loading configuration
Launchpad
Ln 1, Col 1 Spaces: 4 UTF-8 LF JavaScript JSX Finish Setup Go Live Prettier


File Edit Selection View Go Run
E-Grivernece-Hub-master
EXPLORER
OPEN EDITORS
StudentDashboard.jsx E-Grivernece-Hub-...
E-GRIVERNECE-HUB-MASTER
E-Grivernece-Hub-master
frontend
src
pages
Admin
AdminLogin.jsx
AdminProfile.jsx
AdminSignup.jsx
ManageTypes.jsx
PendingGrievances.jsx
Auth
landing
student
FileGrievance.jsx
GrievanceDetails.jsx
MyGrievances.jsx
StudentDashboard.jsx
styles
SuperAdmin
User
OUTLINE
TIMELINE
RUNNING TASKS
StudentDashboard.jsx
E-Grivernece-Hub-master > frontend > src > pages > student > StudentDashboard.jsx > ...
import React, { useEffect, useState } from "react";
// import "..styles/StudentDashboard.css";
import api from "../../api/axiosInstance";
import { useNavigate } from "react-router-dom";
const StudentDashboard = () => {
const [user, setUser] = useState(null);
const [grievances, setGrievances] = useState([]);
const navigate: NavigateFunction
const navigate = useNavigate();
useEffect(() => {
fetchUser();
fetchMyGrievances();
}, []);
// Fetch logged-in user
const fetchUser = async () => {
PROBLEMS OUTPUT
Filter
Pip package updater
Loading configuration....
Done loading configuration
Launchpad
Ln 1, Col 1 Spaces: 4 UTF-8 LF JavaScript JSX Finish Setup Go Live Prettier

Chapter 4

Results & Analysis :

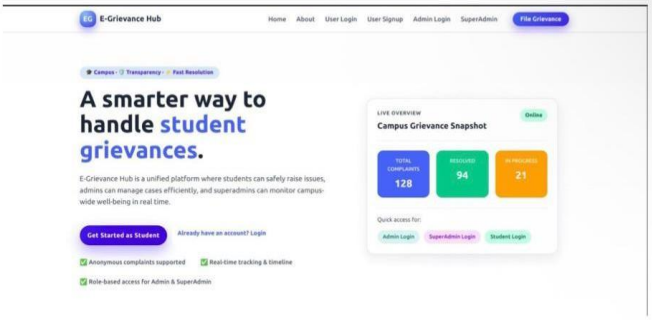
Challenges :

1. One of the biggest challenges was working with the MERN stack as a complete system. Managing frontend components, backend routes, database schema, and authentication together required extra time and understanding.
2. Handling file uploads such as images and PDFs was more difficult than expected. Setting up FormData on the frontend and configuring Multer on the backend needed careful testing to avoid errors and file-handling issues.
3. Implementing role-based access control was another challenge. Each user type— Student, Admin, and SuperAdmin—needed different permissions, dashboards, and API access rules, which made backend logic more complex.
4. Creating a proper grievance timeline was challenging because every status update, comment, or action had to be recorded accurately. Even a small mistake could break the entire tracking history.
5. Setting up email notifications required correct configuration of SMTP settings. Making sure that emails consistently reached the user and did not end up in spam folders also needed extra adjustments.
6. Designing a simple and user-friendly interface took time. The UI had to remain easy to use while supporting multiple features like file uploads, comments, and tracking.
7. Deployment was also challenging. Connecting the frontend, backend, and MongoDB Atlas together required proper environment settings, CORS configuration, and cloud setup.

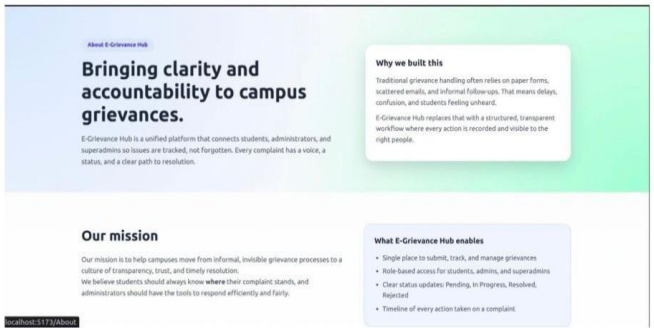
Limitations :

1. The system does not yet include a live chat feature. Communication happens only through comments and notifications.
2. The analytics section for the SuperAdmin is basic. More detailed reports and visual charts can be added in the future.
3. The optional AI-based grievance classification is not fully implemented. A more advanced routing system can improve accuracy.
4. There are file size restrictions in the upload feature to prevent server overload. Very large files cannot be uploaded.
5. The system currently works only as a web application. A mobile app version would make it even more convenient for students.
6. The notification system is functional but can be expanded with push notifications or real-time updates.

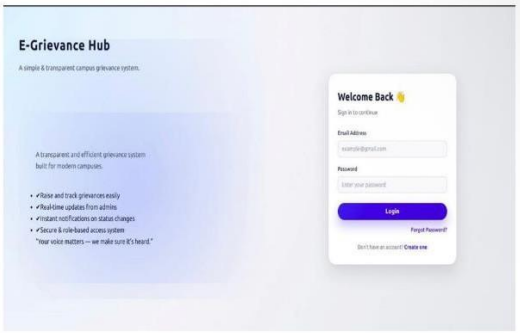
USER INTERFACE



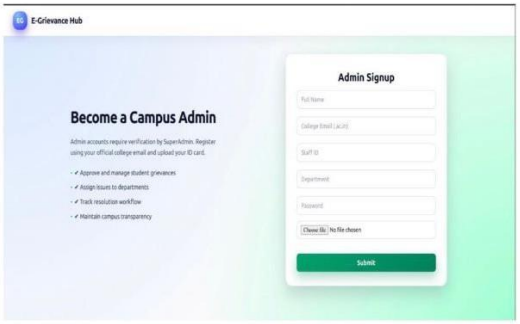
ABOUT



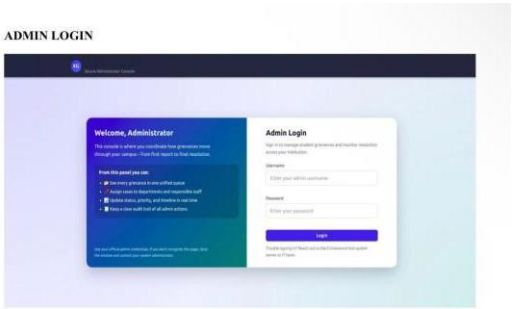
USER LOGIN



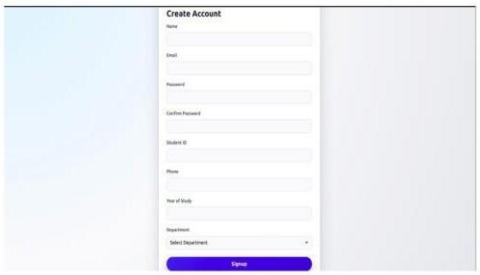
ADMIN SINGUP



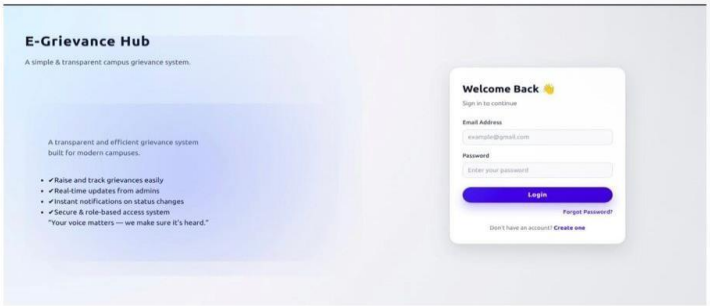
ADMIN LOGIN



USER SIGNUP

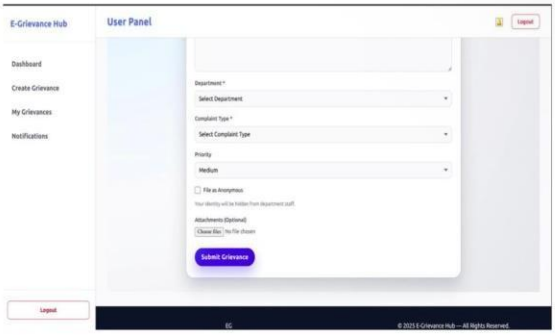


SUPER LOGIN



USER DASHBOARD

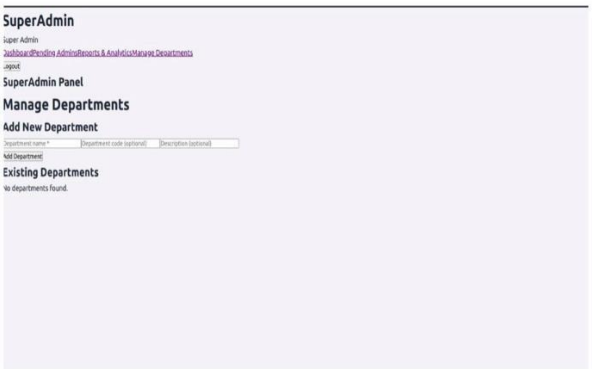
CREATE GRIEVANCE



MY GRIVENCES

SUPERADMIN REPORTS

SUPER ADMIN MANGAE DEPARTMENTS



Chapter 5

Future Scope :

The current version of the E-Grievance Hub covers all the basic requirements of a digital grievance system, but there is still a lot of room for improvement. With more time and resources, the platform can be expanded to make it more powerful, user-friendly, and suitable for large institutions.

The following points highlight the future scope of this project.

1. One of the main improvements planned for the future is adding a real-time chat or messaging feature. This would allow students and administrators to communicate instantly, instead of relying only on comments and notifications.
2. A mobile application version for Android and iOS can be developed. Many students prefer using mobile apps, and having an app would make grievance submission and tracking even more convenient.
3. The system can integrate advanced analytics and visual dashboards for the SuperAdmin. This will help the institution understand common grievance areas, analyze trends, and take better decisions.
4. AI-based grievance classification can be improved further. With machine learning, the system can automatically categorize complaints and route them to the correct department without manual intervention.
5. Cloud storage integration can be added to handle large files and attachments more efficiently. This will also improve the system's performance during peak usage.
6. The platform can support multilingual options so that students from different backgrounds can use the system in their preferred language.
7. The security features can be enhanced by adding multi-factor authentication, activity logs, and better encryption for sensitive data.
8. An automated reminder and escalation system can be added so that unresolved grievances are automatically flagged to higher authorities.
9. A feedback system can be introduced where students can rate the grievance resolution process. This will help the institution improve its services over time. 10.Integration with college ERP or student portals can be added so that the grievance system becomes a part of the institution's main digital ecosystem.

Overall, there are many possibilities to expand this system. With the right improvements, the E-Grievance Hub can become a complete institutional grievance management solution that enhances transparency, accountability, and communication.

Conclusion :

The E-Grievance Hub was developed with the goal of making the grievance process in educational institutions simpler, faster, and more transparent. Throughout the project, I focused on understanding the real problems students face when submitting complaints and the challenges administrators experience while handling them. By moving the entire process online, this system reduces confusion, saves time, and ensures that grievances are not lost or ignored. The platform allows students to submit complaints easily, upload supporting documents, and track every update through a clear timeline. On the other side, administrators can manage grievances in an organized way using their dedicated dashboard. The SuperAdmin features further strengthen the system by ensuring proper monitoring, department management, and admin verification. Working on this project helped me understand how important digital grievance systems are in modern institutions. It also gave me hands-on experience with the MERN stack, authentication methods, file uploads, and real-time communication between the frontend and backend. While the system works effectively, there is still room for improvement, such as adding real-time chat, better analytics, mobile apps, and advanced AI features. Overall, the E-Grievance Hub successfully demonstrates how technology can improve communication and accountability between students and administration. With further development, the platform can become a complete and reliable solution for handling grievances in any educational organization.

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